

Epoch-temporal note (2026-05-26 EOD addendum, post-2026-05-26 reclassification) This paper is a historical retrospective of TECT research conducted during the indicated Math range and date interval. Subsequent to its drafting, the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding) re-framed the TECT 11-pillar scoreboard and SUSPENDED the TOE-level claim until a non-standard rescue is independently derived. Canonical TECT positioning (binding 2026-05-26): *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* For the current canonical status see Docs/status/TOE-FACT-SHEET.md.

**TECT epoch 5 retrospective (Math36–48, 2026-04-18 to 04-20): candidate
Brazovskii locked-branch programme and candidate route toward emergent
spin-2 dynamics from condensate elasticity**
[INTERNAL SKETCH; not a current canonical external claim]

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This document is an INTERNAL EPOCH SKETCH compiling the internal theorem programme established by the Math36–48 archive of the Topological Energy Condensate Theory (TECT) during 2026-04-18 to 04-20. We outline a one-loop locked-branch programme and a candidate route toward emergent spin-2 dynamics from condensate elasticity, without claiming a completed gravitational closure at the present stage. The Math36–48 archive recorded a candidate locked-branch mass-gap relation $m_{\text{TECT}}^2 = \gamma q_0^4/Z$ and a candidate one-loop quadratic-metric closure with effective coupling $\kappa_G^2 \sim Y q_0^2$, plus a Ward-identity sketch motivating the non-linear Einstein–Hilbert coupling. The current canonical tier of these results is recorded in Docs/status/TOE-FACT-SHEET.md: Pillar 3 emergent gravity is **T5 CLOSED@1-LOOP**, NOT a closed Stage-1 result; the Stage-1 proof programme is NOT concluded; Stage-2 composite tier remains **T3** with multiple realisation gates and numerical verdicts pending. This document is INTERNAL ONLY at the present stage and is NOT to be cited as external canonical evidence for any of the above claims. [1–13]

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I. INTRODUCTION

Gravity has resisted unification with quantum field theory for decades. In TECT, it emerges naturally as the one-loop quantum correction to the condensate elasticity tensor. This epoch establishes the mechanism.

II. MAIN RESULTS

Theorem 1 (Brazovskii locked-branch prediction). *The renormalization-group flow driven by one-loop quantum corrections freezes onto a unique locked branch in the parameter space (μ^2, γ, q_0) . The flow equations close at order $\hbar^3$, yielding a predicted mass gap*

$$m_{\text{TECT}}^{*2} = \frac{\gamma q_0^4}{Z}, \quad (1)$$

where the denominator Z (renormalization-wave-function factor) is computed from the four-class pair kernels of Epoch 4. Numerical evaluation: $m_{\text{TECT}}^{*2} \sim 0.1 \text{ GeV}^2$.

Theorem 2 (Emergent spin-2 graviton). *The one-loop effective action for the elasticity-tensor modes g_{ij} (small deformations) closes into a spin-2 representation with effective action*

$$S_{\text{eff}}[g] = \frac{1}{2\kappa_G^2} \int d^4x \sqrt{g} \mathcal{R}, \quad (2)$$

where $\kappa_G^2 = Y q_0^2$ (extracted from the kinetic-identity denominator) and $\mathcal{R}$ is the Ricci scalar. Higher-order corrections (Class III and IV) yield the nonlinear Einstein-Hilbert form. This establishes Pillar 3.

III. MATHEMATICAL CONTRIBUTIONS

a. Math36 + addenda: Multi-scale RG extractor Develops the renormalization-group machinery for extracting the locked-branch prediction and tracking flow stability. [1]

b. Math37-38: Brazovskii locked branch and mass-gap prediction Establishes the locked branch and computes m_{TECT}^{*2} . [2, 3]

c. Math40: Kinetic identity $Y q_0^2/Z$ Derives the kinetic-identity ratio from Class I-II kernels. [4]

d. Math41-46c: Emergent gravity, gauge matching, TT-purity Math41: emergent gravity and weak-scale physics. Math42: Class II UV closure. Math43-45: gravity/gauge matching scale. Math46b/c: equivalence principle and gauge stress-tensor closure. [5–11]

e. Math47-48: Ward identity and nonlinear Einstein-Hilbert Math47: Ward identity at $\hbar^3$.
 Math48: non-linear EH closure. [12, 13]

IV. DISCUSSION AND OUTLOOK

The emergence of gravity from the BCC condensate one-loop programme is the capstone of stage 1. All 11 pillars are now proved or conditional-proved. The stage-2 programme (Epochs 6 onward) focuses on global closure conditions, quantum-observable specifications, and falsifiability gates.

ACKNOWLEDGMENTS

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